nature

Epigenetic regulation during cancer transitions across 11 tumour types

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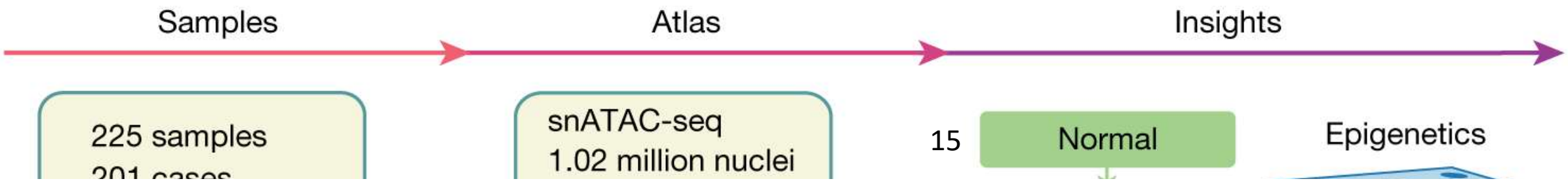
Presenter: Yilu Wang
13.03.2026

What epigenetic changes drive cancer transitions?

- **Chromatin accessibility** regulates gene expression and cellular identity
  - Alterations have been indicated in driving cancer transitions, from normal → primary → metastasis

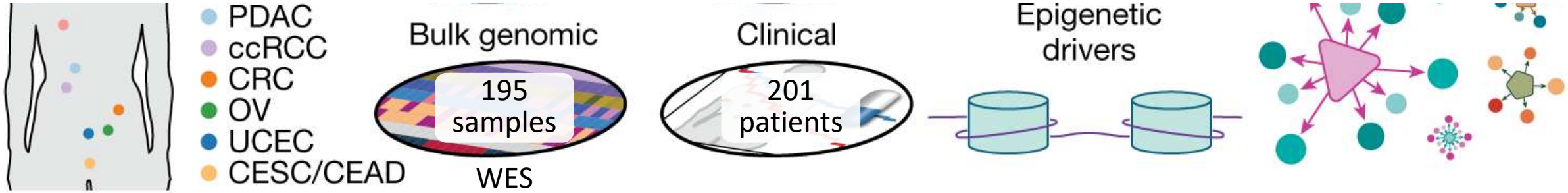
- **Epigenetic drivers** remain less understood than genetic drivers
  - May explain previously unknown tumorigenic mechanisms

Data collection and study design



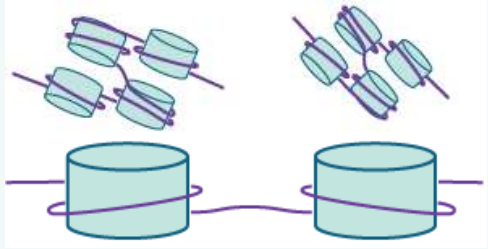
- **Problem:** cancer transitions are epigenetically regulated but poorly mapped at single-cell resolution

- **Approach:** integrated single-cell chromatin and transcriptomic atlas



sn: single-nucleus
sc: single-cell
WES: whole-exome sequencing

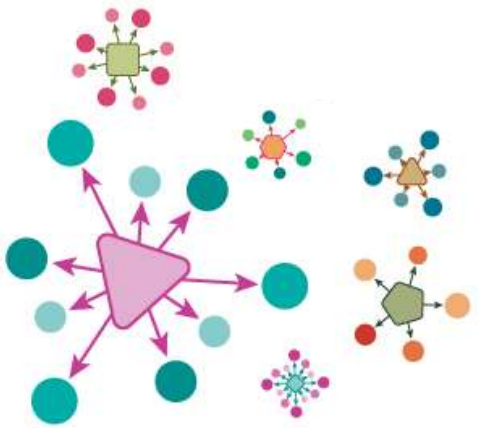
1. Epigenetic changes



DACR
differentially accessible chromatin region

2. Gene expression

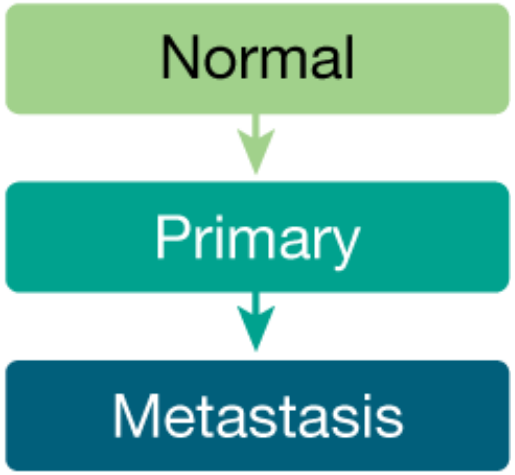
3. TF programs



Regulon
a TF and its target genes

Associations

Cancer transitions



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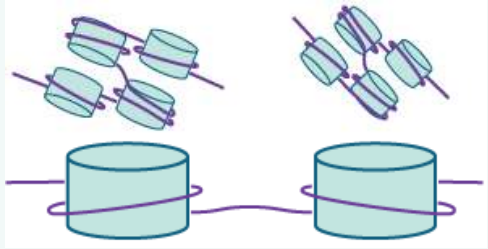

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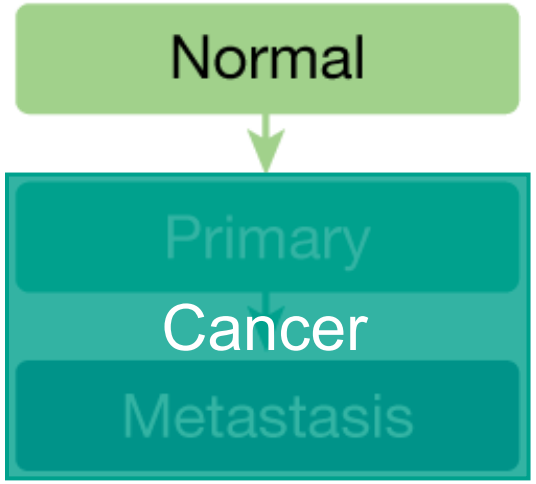
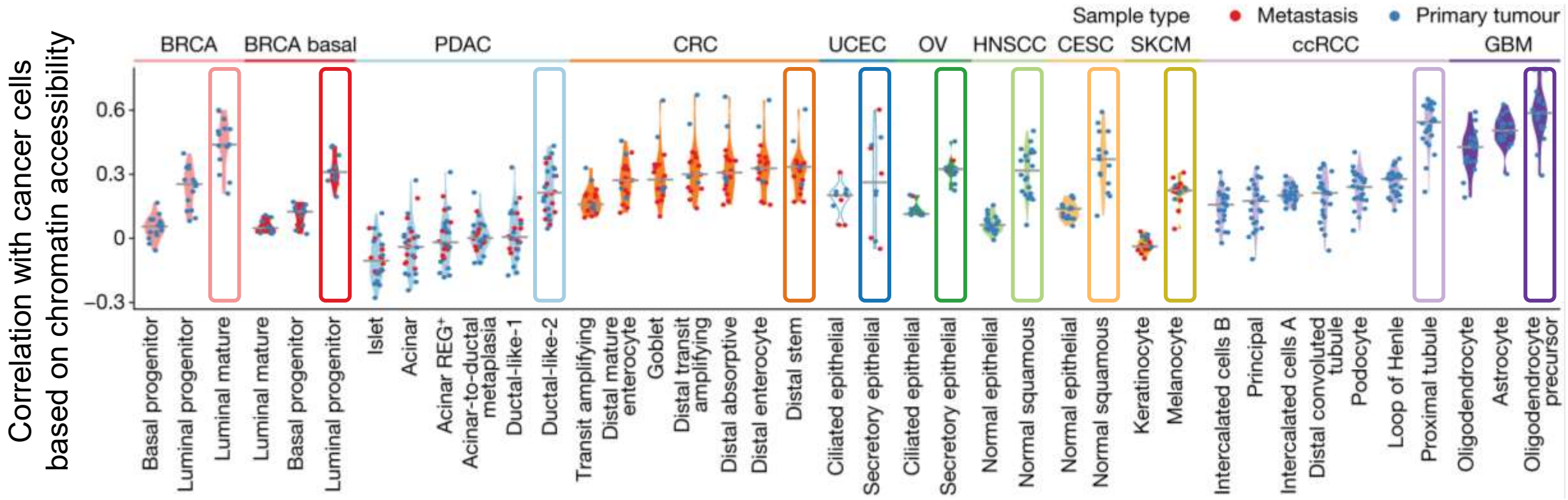
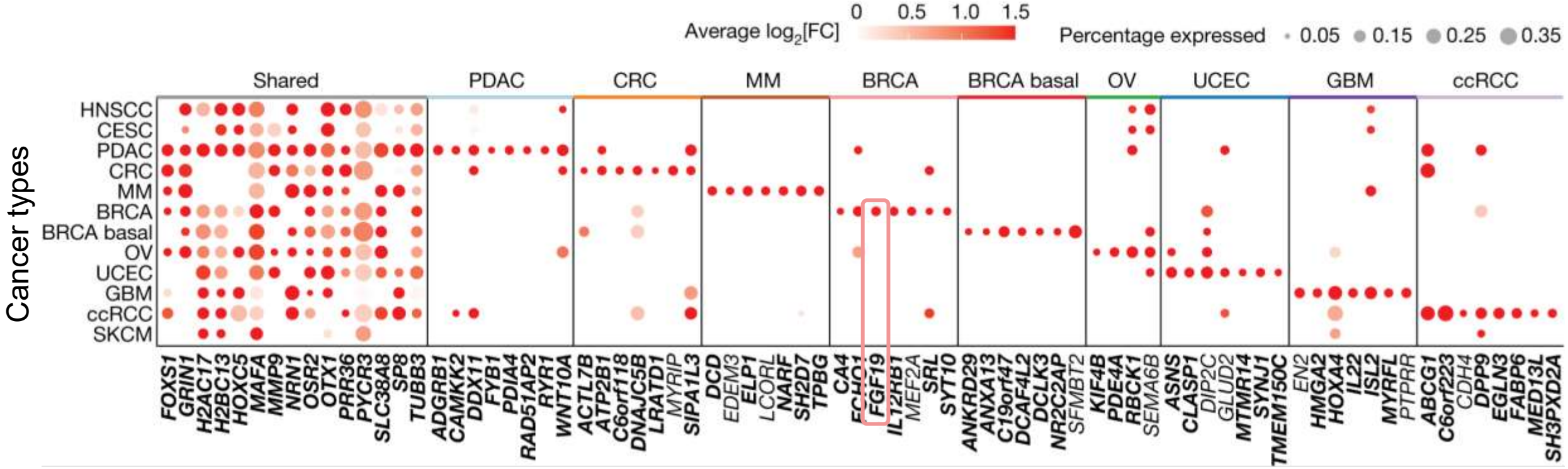


Closest normal cells (CNCs) serve as the reference for cancer cells



Cancer-associated changes in chromatin accessibility can be shared or cancer-type-specific

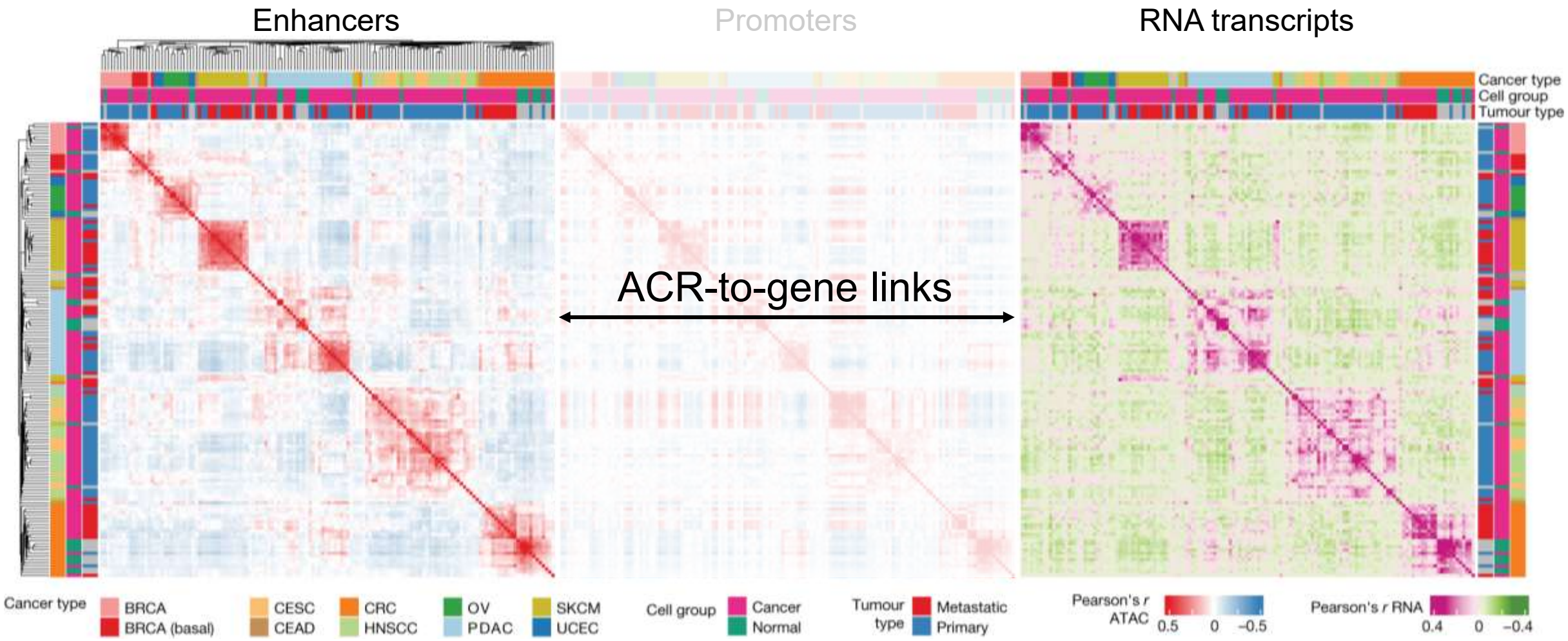


The nearest gene of each DACR

- **Color:** $\log_2 FC = \log_2 \frac{mean\ accessibility\ in\ cancer\ cells}{mean\ accessibility\ in\ CNCs}$
- **Bubble size:** percentage of cancer cells with the accessible DACR
- **Bold:** genes whose DACRs overlap with cis-regulatory elements (promoters or enhancers) from the EpiMap annotation

DACR: differentially accessible chromatin regions

Enhancer accessibility better reflects gene expression and is more cancer-type-specific

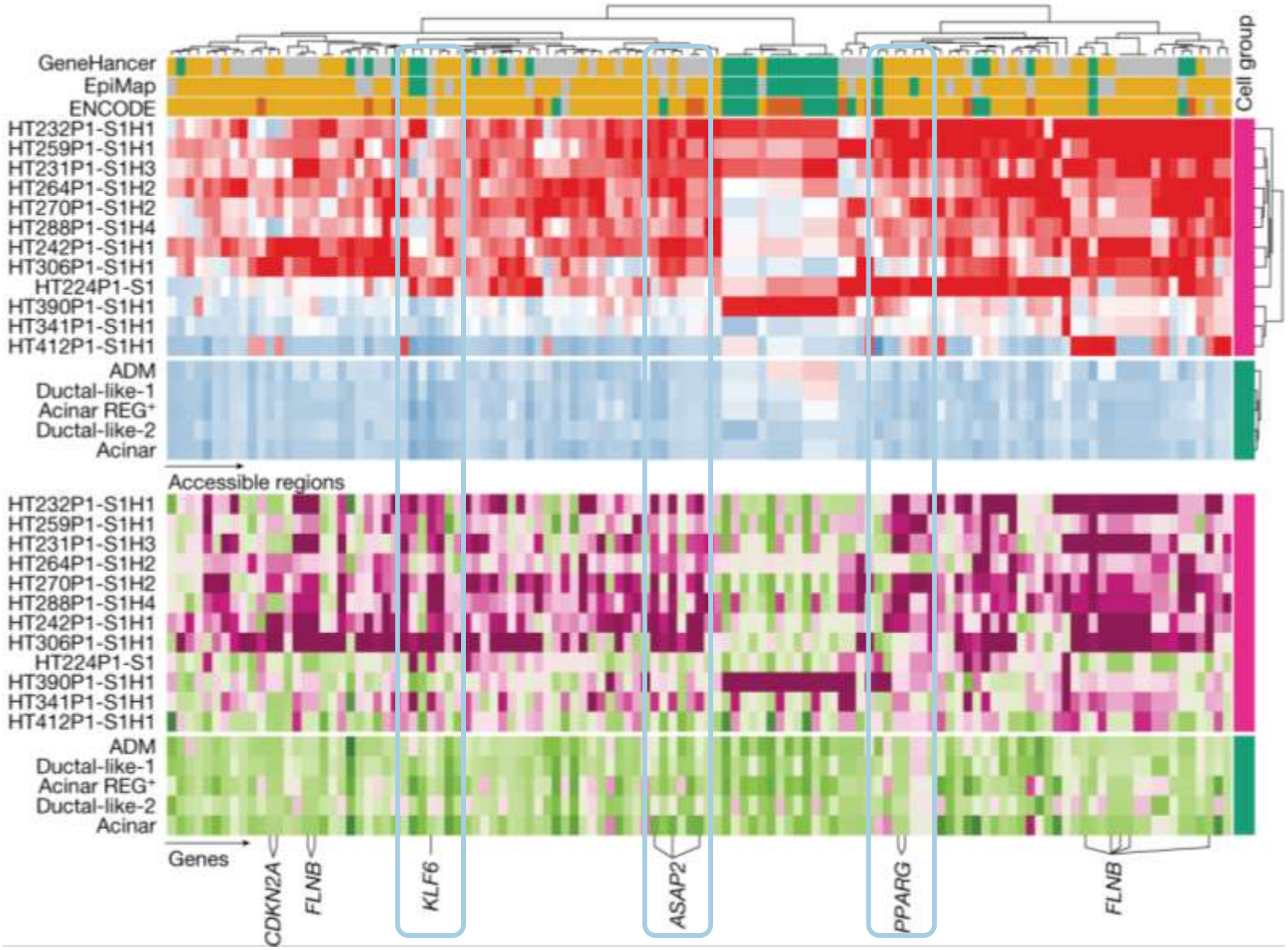


ACR: accessible chromatin regions

PDAC example: ACR-to-gene links reveal candidate regulatory elements

ACR accessibility

Gene expression



Cancer cell samples

Normal cell types

Cancer cell samples

Normal cell types

PDAC: pancreatic ductal adenocarcinoma

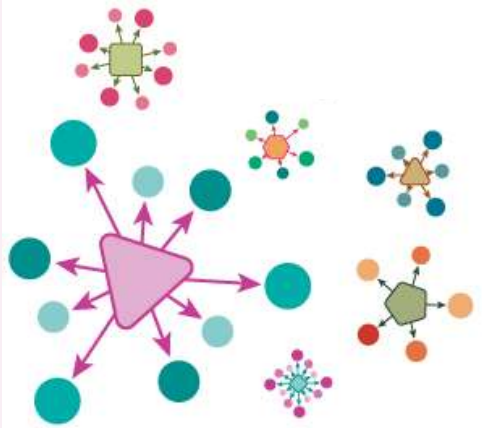
1. Epigenetic drivers

DACR

differentially accessible chromatin region

2. Gene expression

3. TF programs



Regulon

a TF and its target genes

Associations

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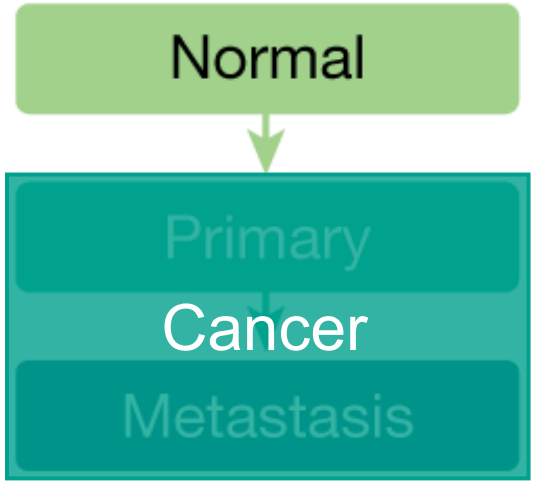
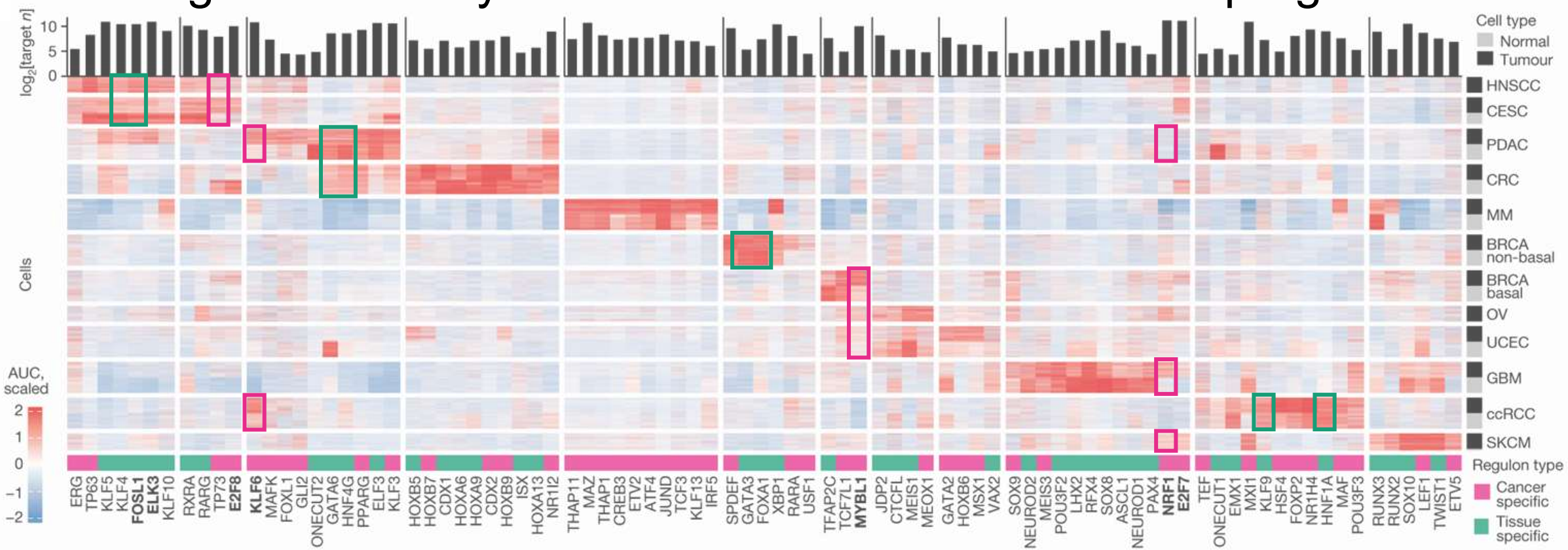
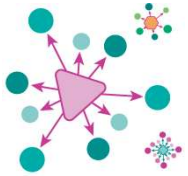


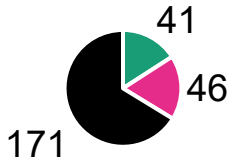
Regulons identify candidate cancer-associated TF programs



258 regulons with concordant expression between the TF and its target genes:



Identified by *SCENIC*

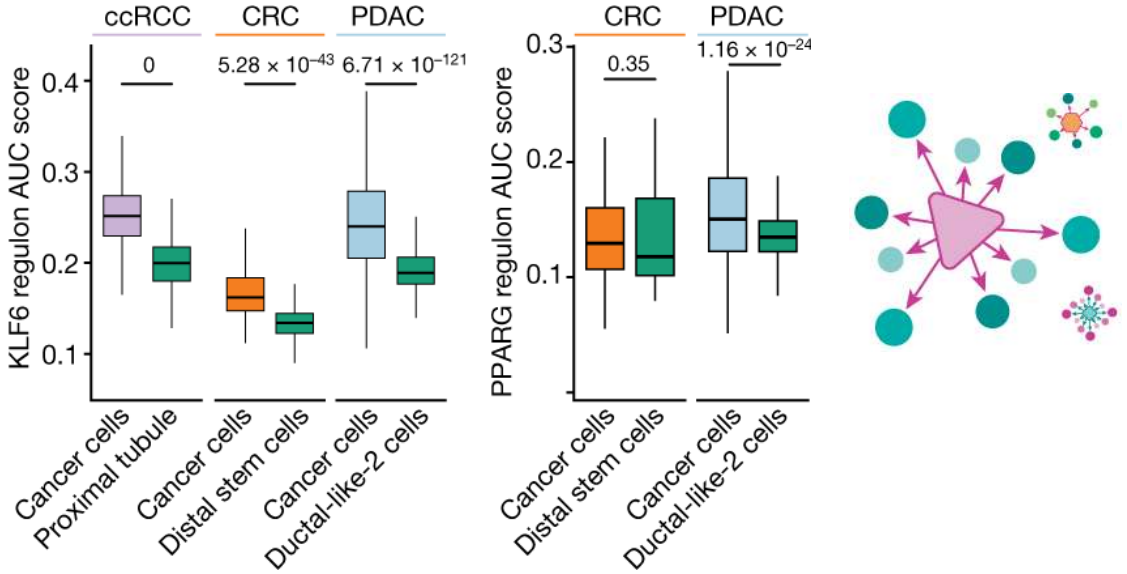


■ Tissue-specific ■ Cancer-cell-specific

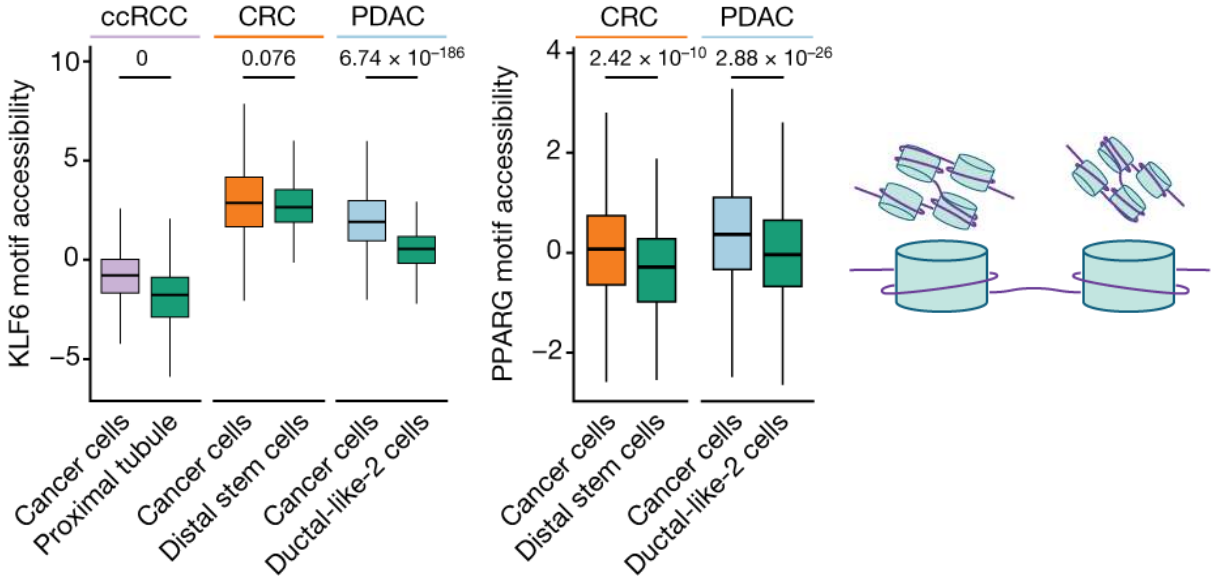
Examples: cancer-associated regulons show concordant TF motif accessibility

KLF6 PPARG

Regulon activity

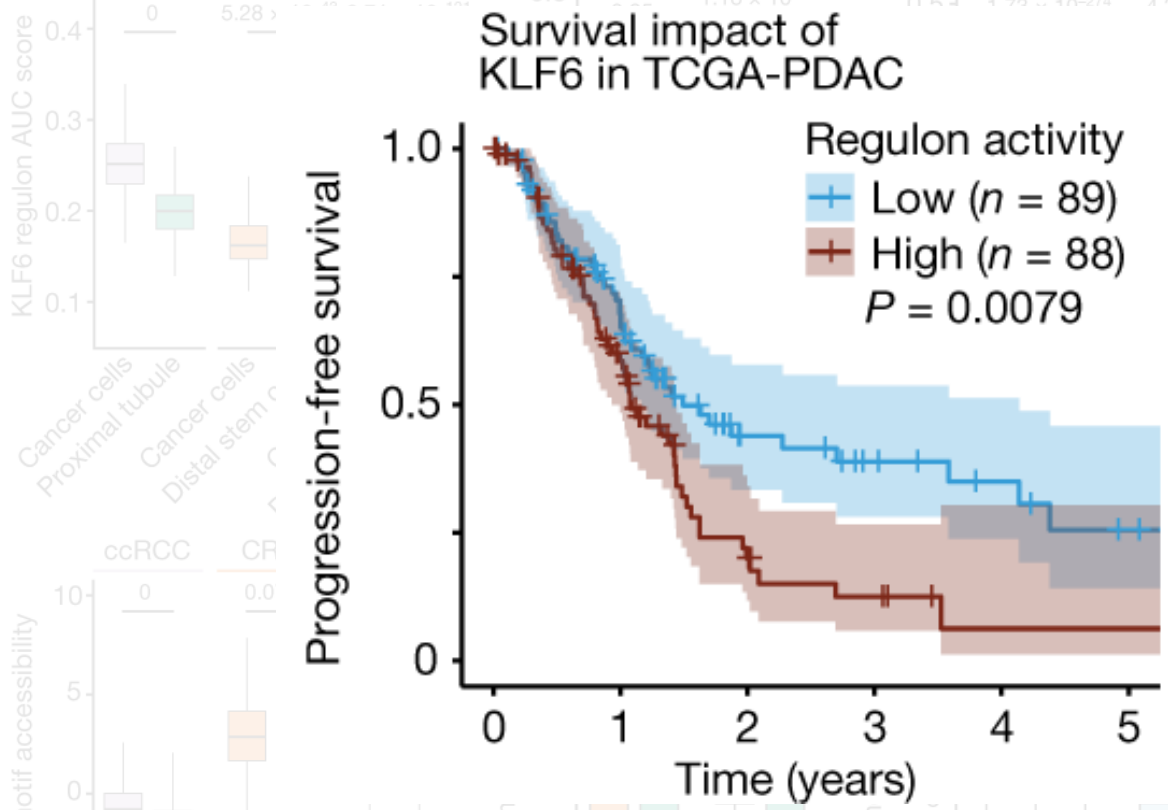


TF motif accessibility



Cancer-cell-specific regulons show concordant TF motif accessibility

KLF6 regulon activity stratify PDAC patient survival



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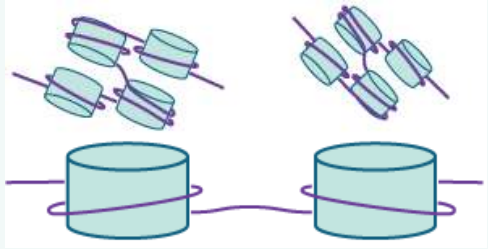
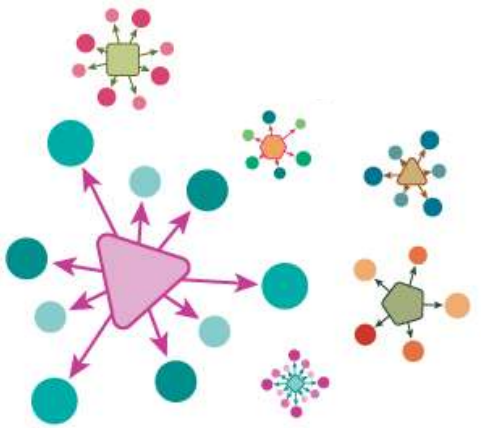


DACR
differentially accessible chromatin region

2. Gene expression

3. TF programs



Regulon
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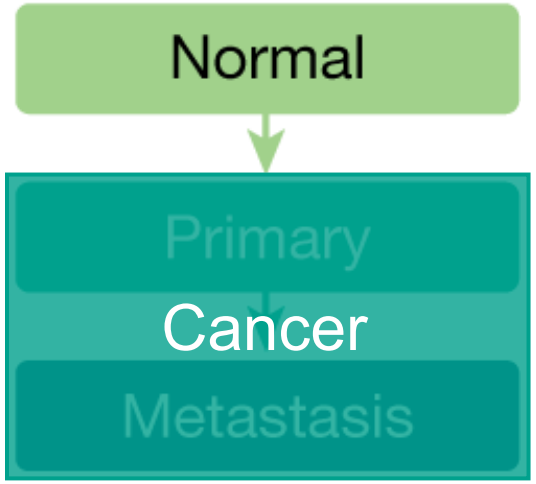
Associations

Cancer transitions



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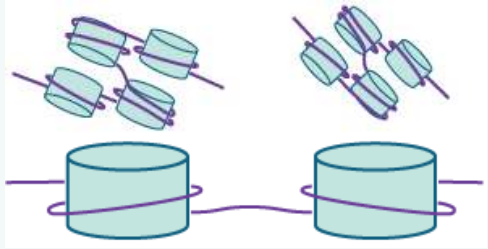

DACR
differentially accessible chromatin region

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Associations

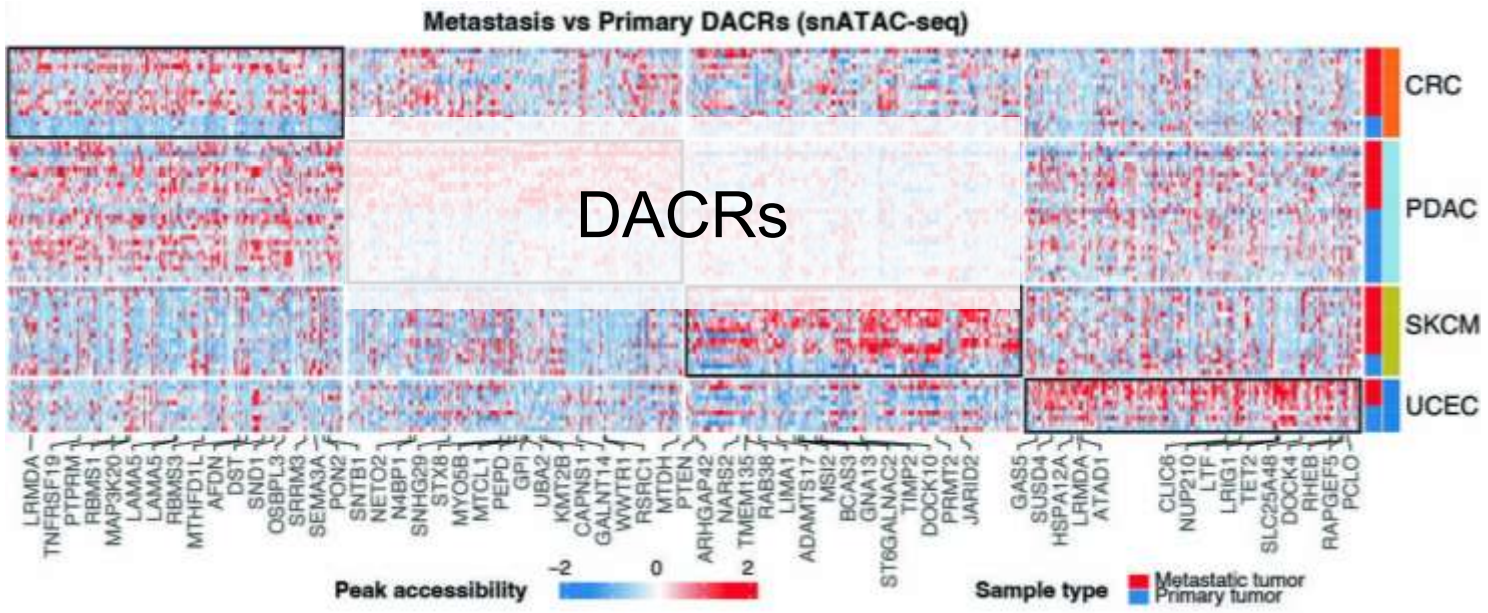
Cancer transitions

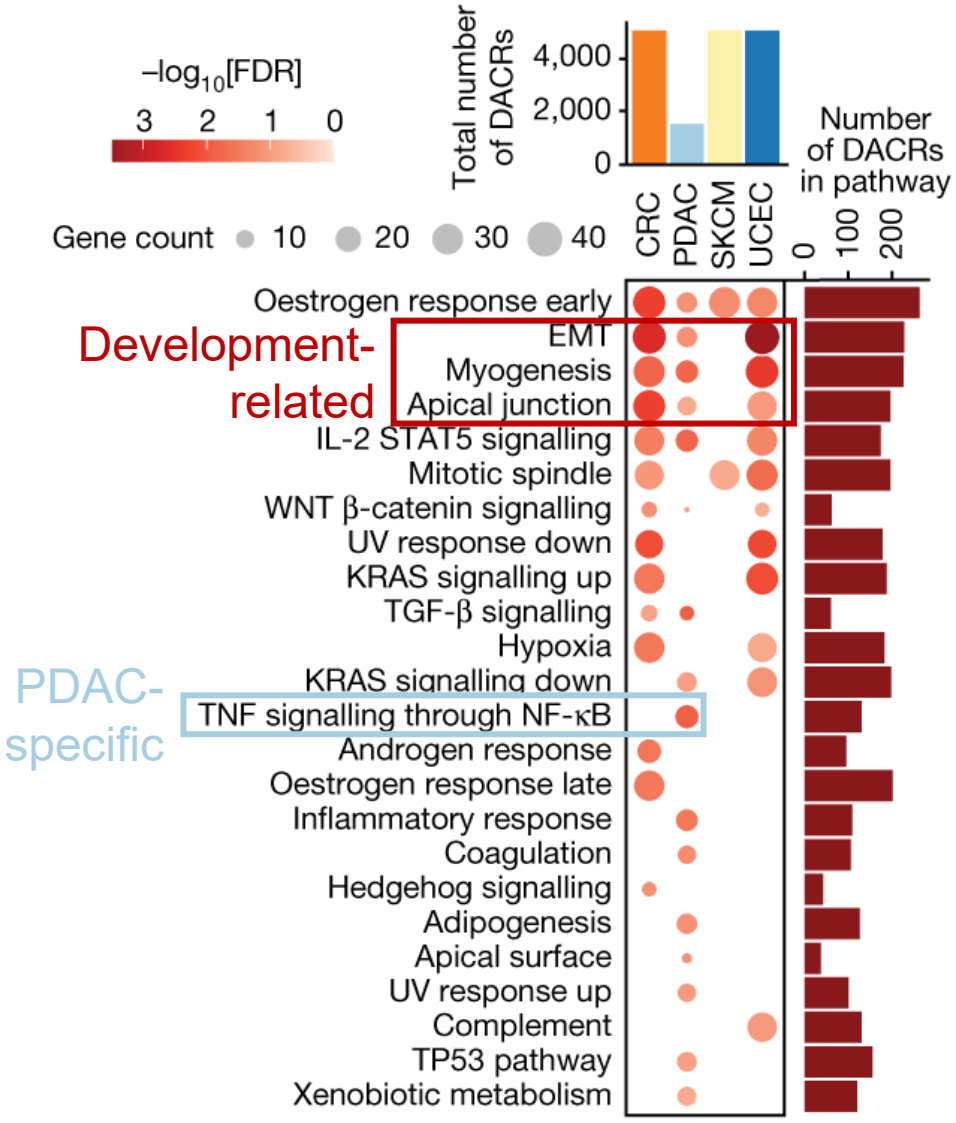
Normal

Primary

Metastasis

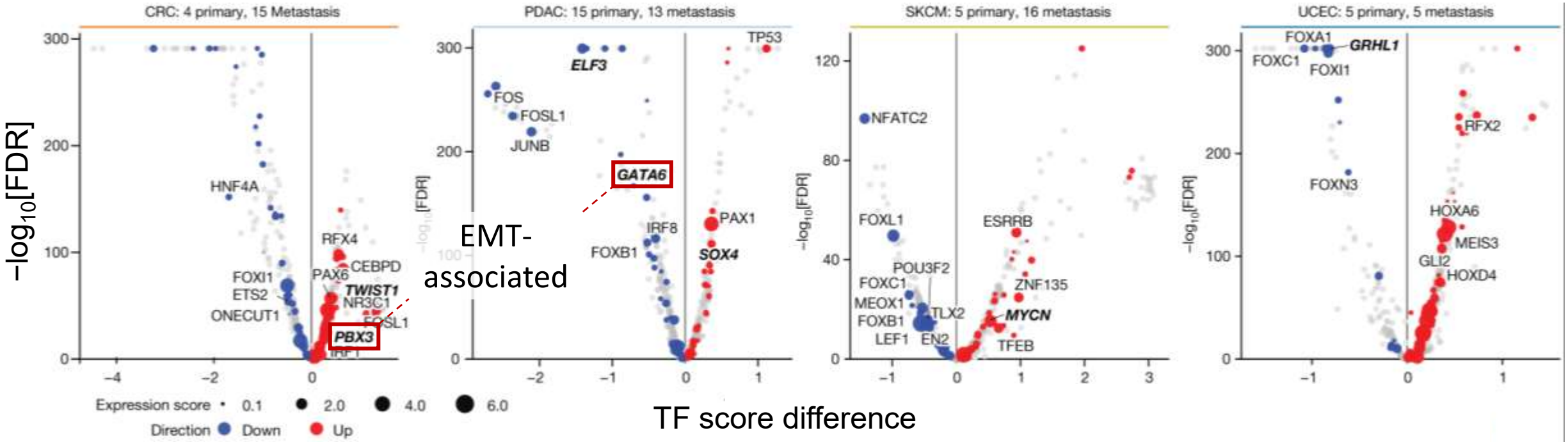
In four cancers, metastasis-associated changes in chromatin accessibility are enriched in certain pathways





DACR: differentially accessible chromatin regions
EMT: Epithelial-to-Mesenchymal Transition

In four cancers, metastasis is associated with cancer-type-specific changes in TF motif accessibility and expression

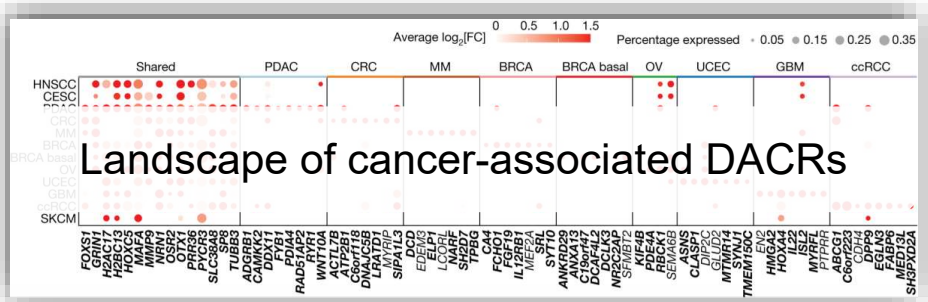


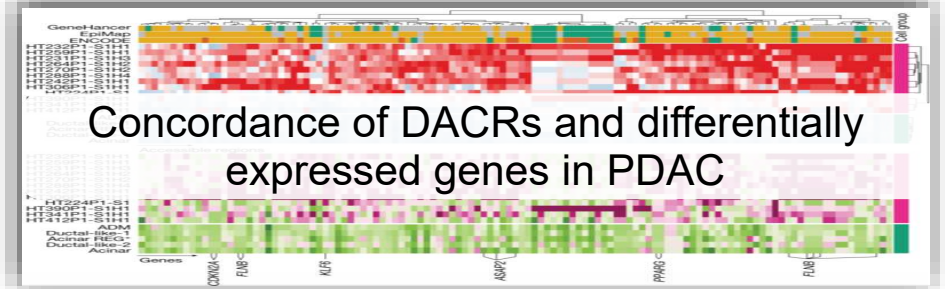
- **X-axis:** TF score difference representing motif accessibility change
- **Dot size:** expression score = $|\log_2 FC| = |\log_2 \frac{TF\ expression\ of\ metastasis\ samples}{TF\ expression\ of\ primary\ samples}|$

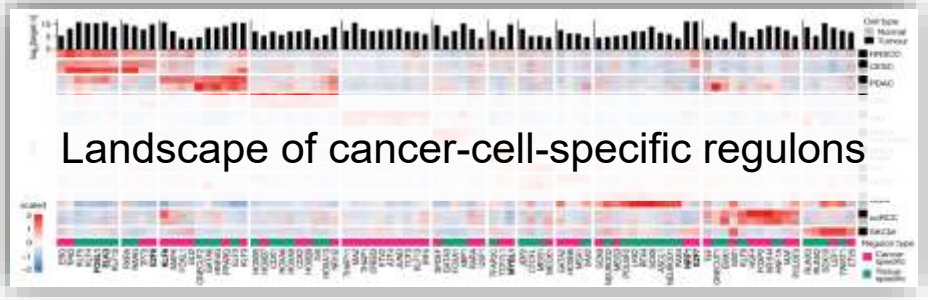
EMT: Epithelial-to-Mesenchymal Transition

What epigenetic changes drive cancer transitions?

What epigenetic changes drive tumorigenesis?





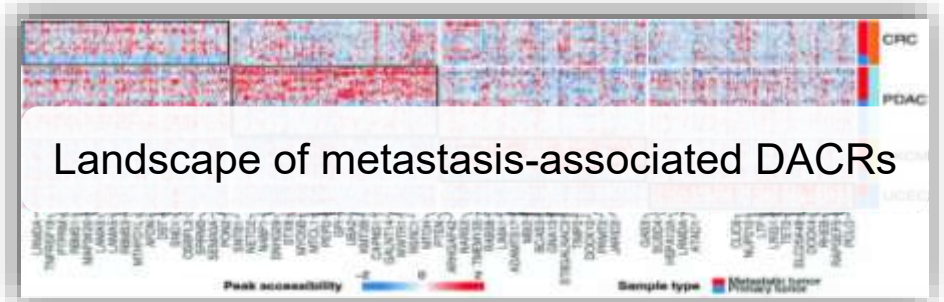


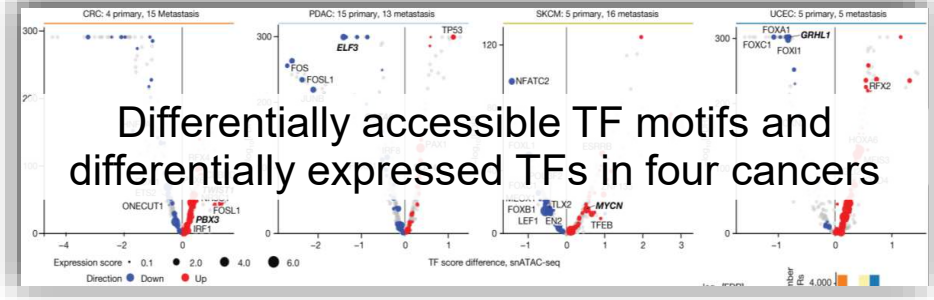
snATAC-seq

snMultiome-seq

sc/snRNA-seq

What epigenetic changes drive cancer metastasis?





Limitation: descriptive atlas vs mechanistic claims

Lacking functional validation

- Every highlighted finding relies on correlation
- Should use knockdowns, CRISPR screens, or perturbation experiments to establish causality

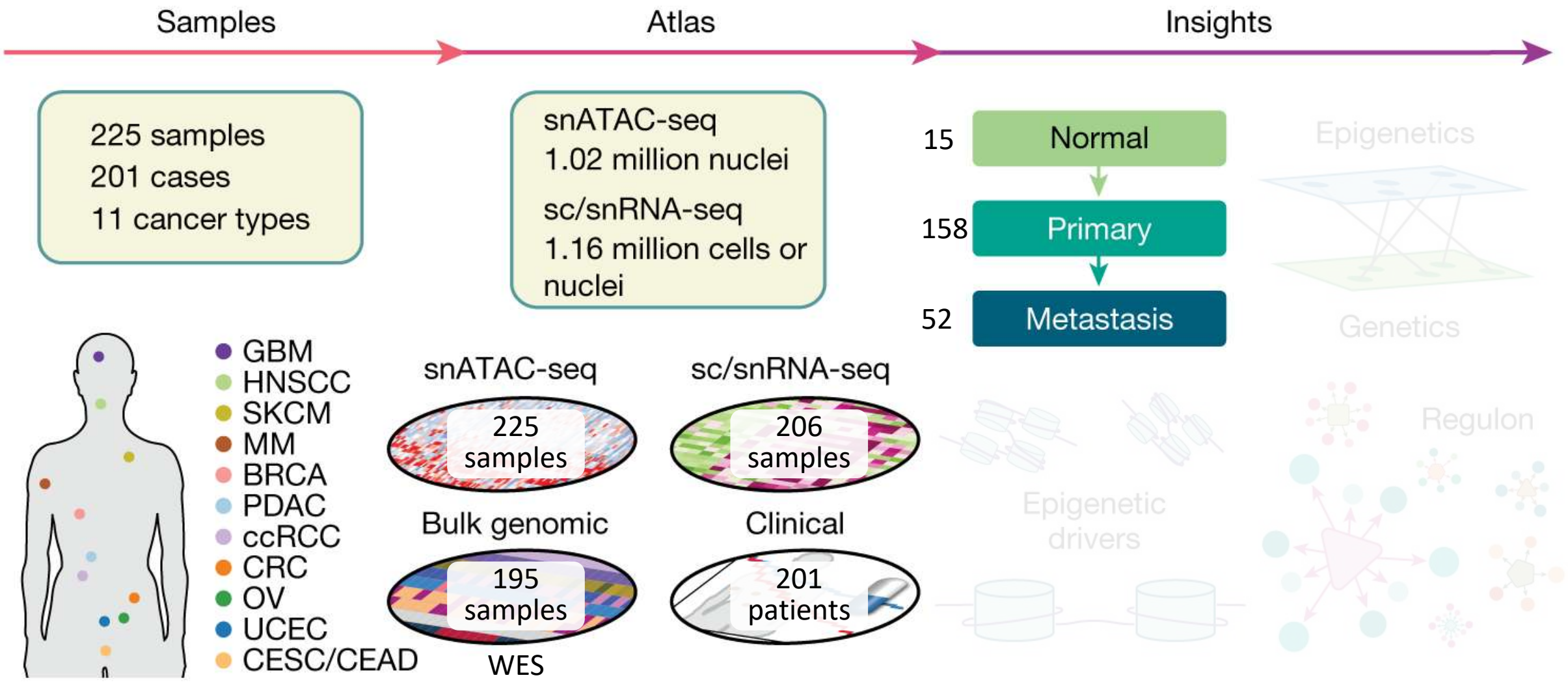
Statistical ≠ biological significance

- With over a million cells, even minimal effect sizes achieve significance
- Independent validation cohorts or predefined effect size thresholds would help distinguish discoveries

Other concerns

- Unbalanced metastasis cohorts limited to four cancer types
- Literature-guided hit selection over unbiased discovery
- Sensitivity of all downstream results to the choice of closest normal cells

Contribution: a broad hypothesis-generating atlas



Thank you!

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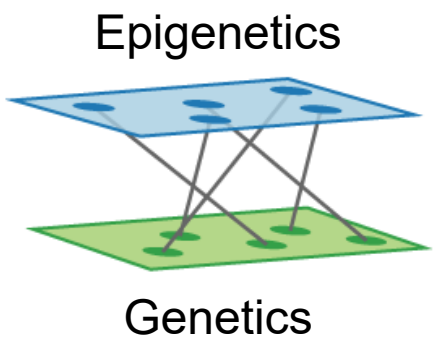
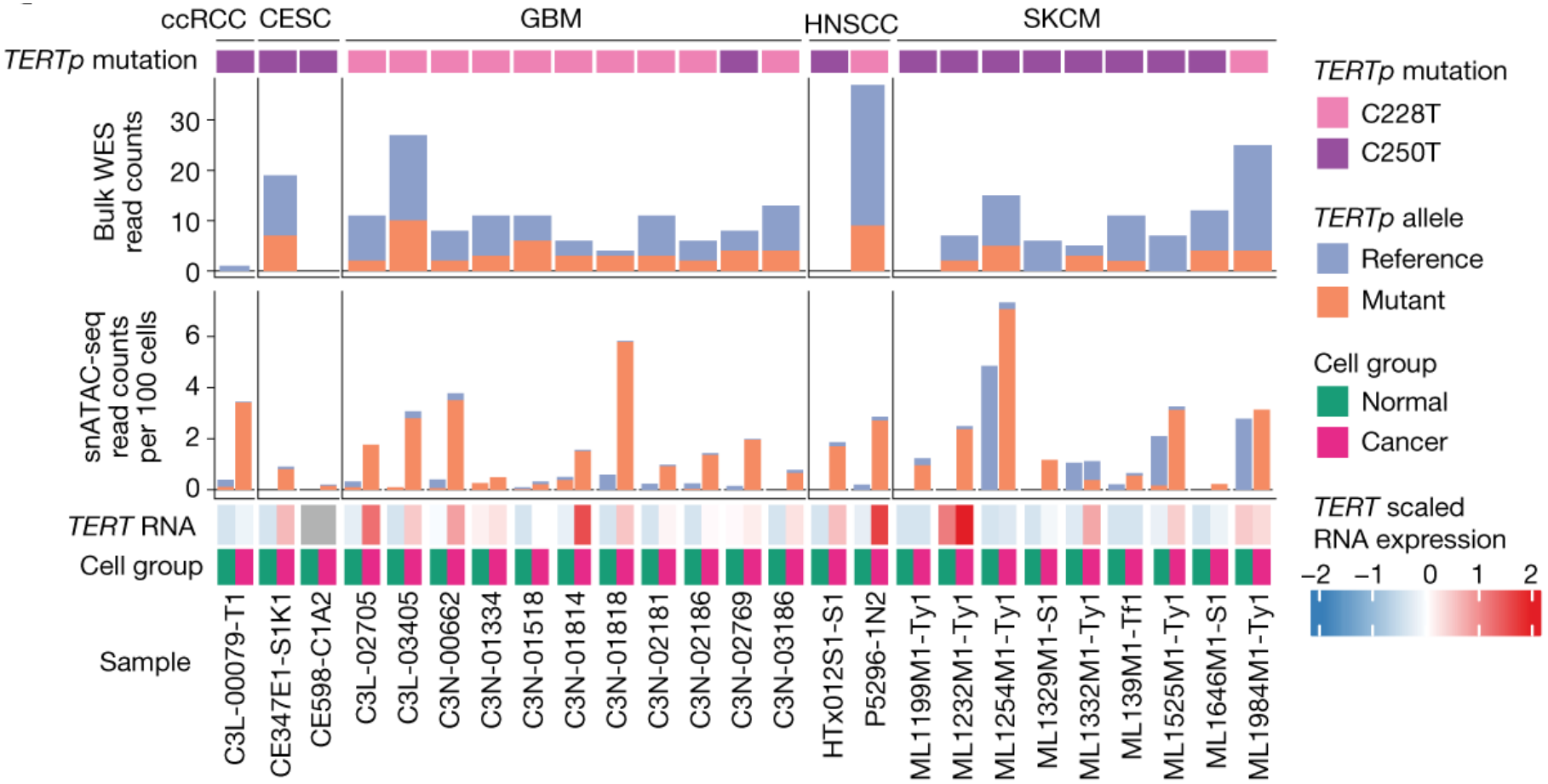
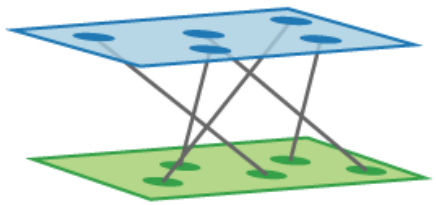
How do epigenetic drivers interact with genetic drivers?

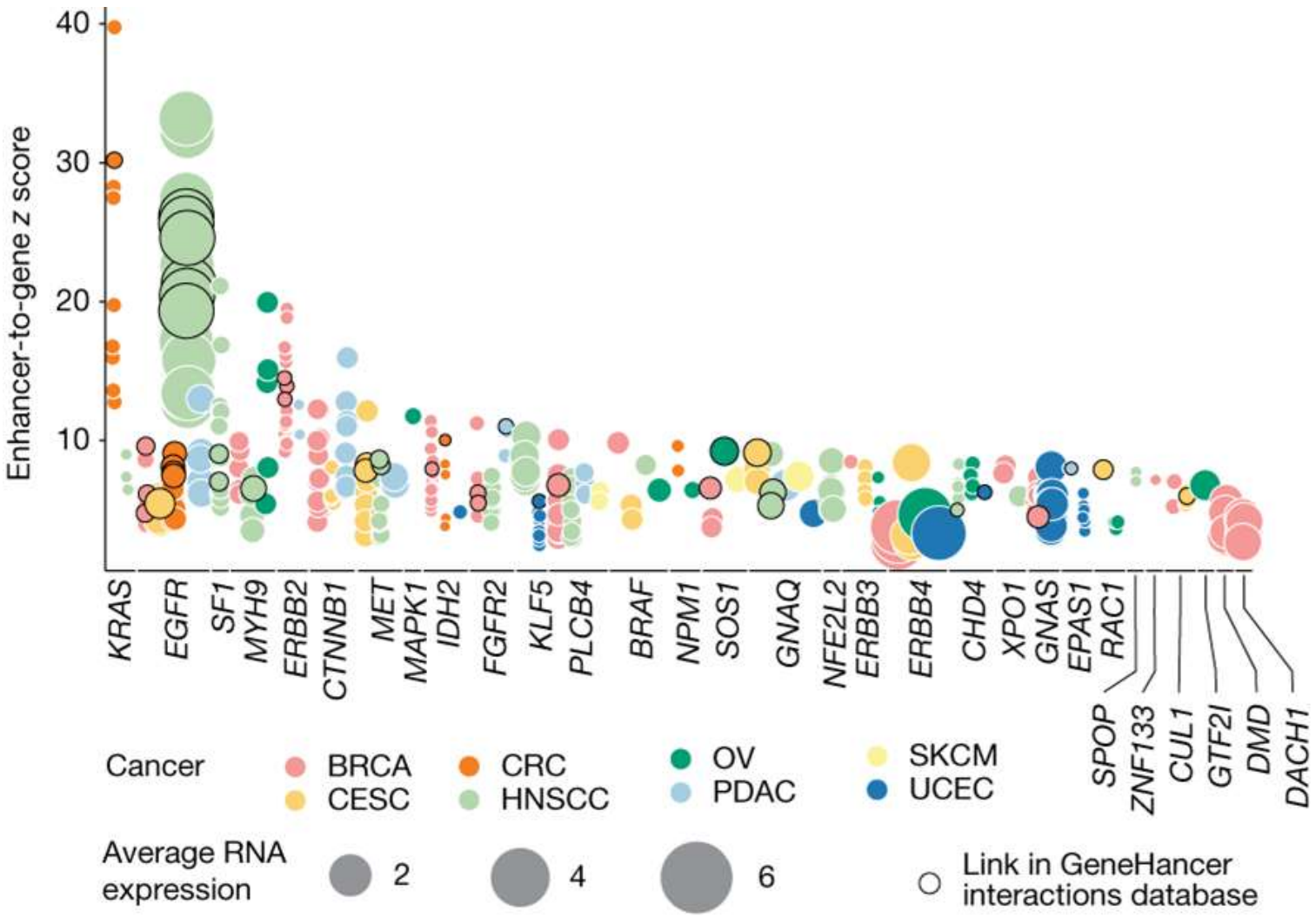
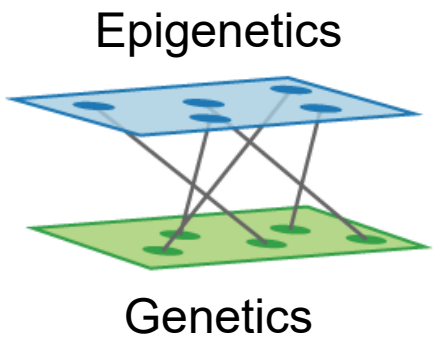


Promoter example: TERTp mutations are associated with increased accessibility in cancer cells



Epigenetics



Genetics

Pan-cancer enhancer-to-gene links identify candidate regulatory elements for recurrent oncogenes